

Unit I :: Distribution of Velocities and Molecular Collisions

☞ **PHY0500304: Heat & Thermodynamics** ☞

Dr. Parag Bhattacharya
Department of Physics
Rangapara College (Autonomous)

Syllabus

Unit I: Distribution of Velocities and Molecular Collisions

Maxwell-Boltzmann Law of Distribution of Velocities in an Ideal Gas and its Experimental Verification. Mean, RMS and Most Probable Speeds. Degrees of Freedom. Law of Equipartition of Energy (No proof required). Mean Free Path. Collision Probability. Transport Phenomenon in Ideal Gases: (1) Viscosity, and (2) Thermal Conductivity. Brownian Motion (qualitative idea only).

1 Maxwell-Boltzmann distribution law

1.1 Postulates of kinetic theory of gases

1.2 Fundamental assumptions

1.3 Velocity space

1.4 Derivation of Maxwell-Boltzmann distribution law

1.4.1 Molecular distribution of speeds

1.4.2 Determination of the constants

2 Molecular speeds

2.1 Average or mean speed

2.2 Root mean squared (RMS) speed

2.3 Most probable speed

3 Energy distribution of a Maxwellian gas

Insert section content.

4 Molecular energies

4.1 Average energy

4.2 Most probable energy

5 Molecular collisions

5.1 Mean free path